Image Classification

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Image Classification

- **What is the image classification task, and what are its application scenarios?**
- What are the challenges in image classification tasks?
- Is a rule-based method feasible?
- What is the data-driven image classification paradigm?
- What are the commonly used evaluation metrics for classification tasks?

What is an image classification task?

- The image classification task is a core task in computer vision, aiming to distinguish images of different categories based on the different features reflected in the image information.

What is an Image Classification Task?

Image Classification: The task of assigning a category label from a fixed set of labels to an input image.

Labels: {Dog, Cat, Truck, Plane, …}

What are its applications?

Jade Vine is a rare flower found only in the Philippine rainforest.

Its green and blue hues earn it the name "Jade Vine".

Due to habitat loss, it is now an endangered species that can only be cultivated in specific soils.

What are its applications?

The Bedlington Terrier originated in Great Britain. Its primary purpose was to hunt foxes, hares, and badgers. From the late 18th to the early 19th century, it was developed by crossbreeding breeds like the Whippet and Dandie Dinmont Terrier. Through selective breeding, it has evolved into the current height, beauty, speed, and agility, while retaining its original vitality and endurance.

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Semantic Gap

- Crossing the "Semantic Gap": Establishing Mappings from Pixels to Semantics

What We See

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What the Machine Sees

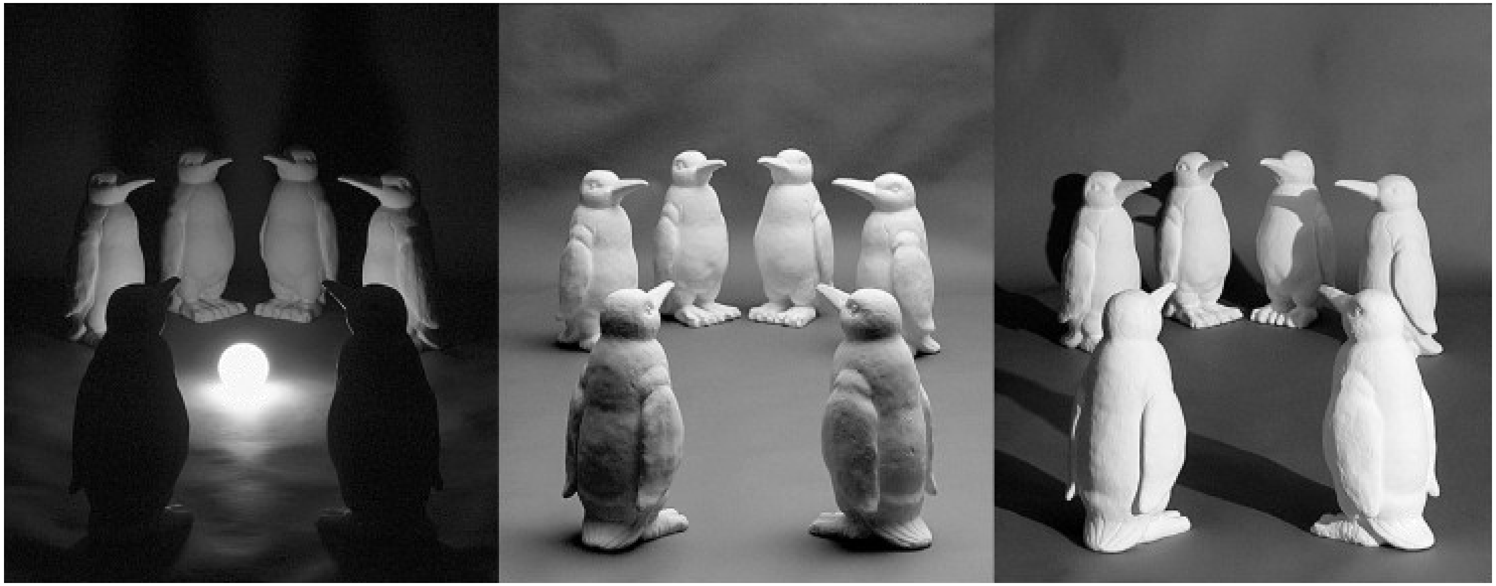
Source: S. Narasimhan

Perspective

Michelangelo 1475-1564

Source: S. Lazebnik

Lighting



Source: J. Koenderink

Scale

Source: S. Lazebnik

Occlusion

Source： Fei-Fei Li

Deformation

Source： Fei-Fei Li

Background Clutter

Intraclass Deformation






Motion Blur

Source: S. Lazebnik

A wide variety of categories

Source：Fei-Fei Li

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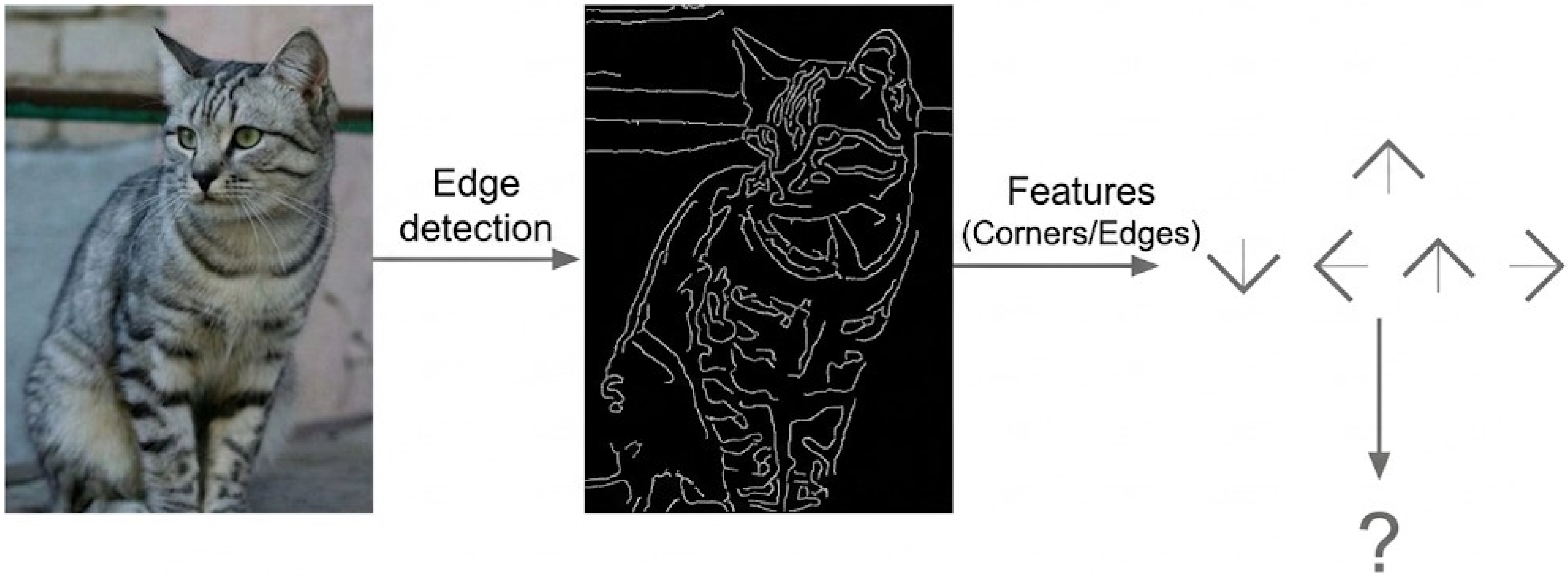
Is a Rule-Based Classification Method Feasible?

```
def classify_image ( image ):
    # Do something magical here
    return class_label
```

Identifing cats or other classes using hard-coding methods

Is a Rule-Based Classification Method Feasible?

Previous attempts:



Is Rule-Based Classification Feasible?

Using hard-coding to identify cats or other classes is a very difficult task.

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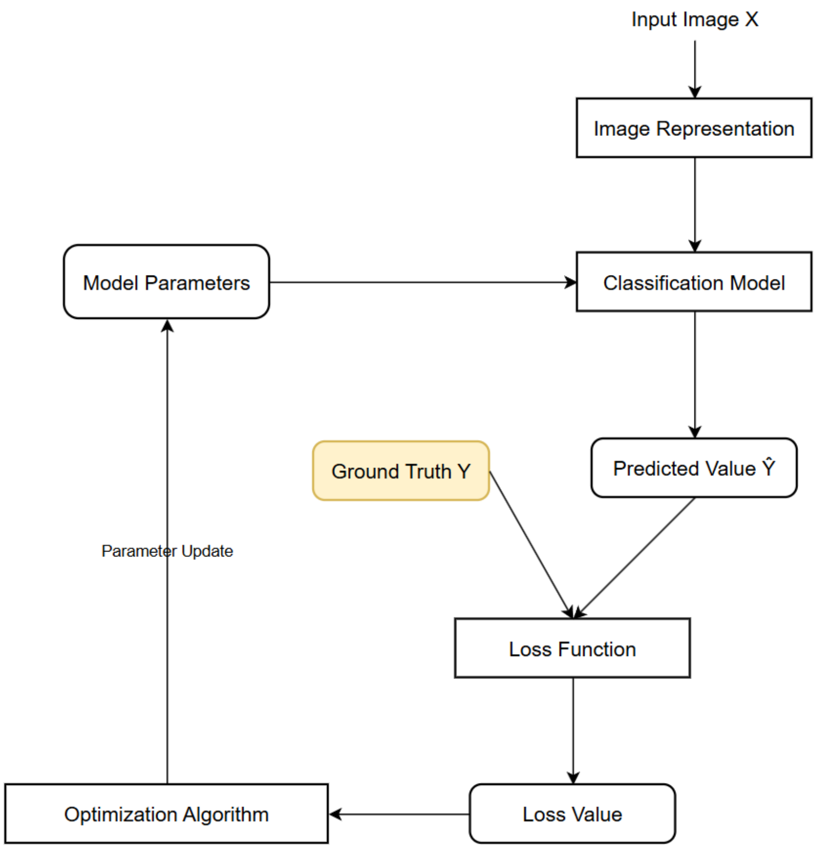
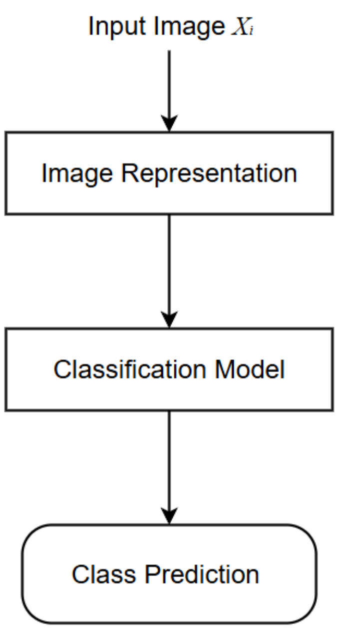
Data-Driven Image Classification Method

1. Dataset Construction
2. Classifier Design and Learning
3. Classifier Decision-Making

Classifier Design and Learning





Classifier Design and Learning

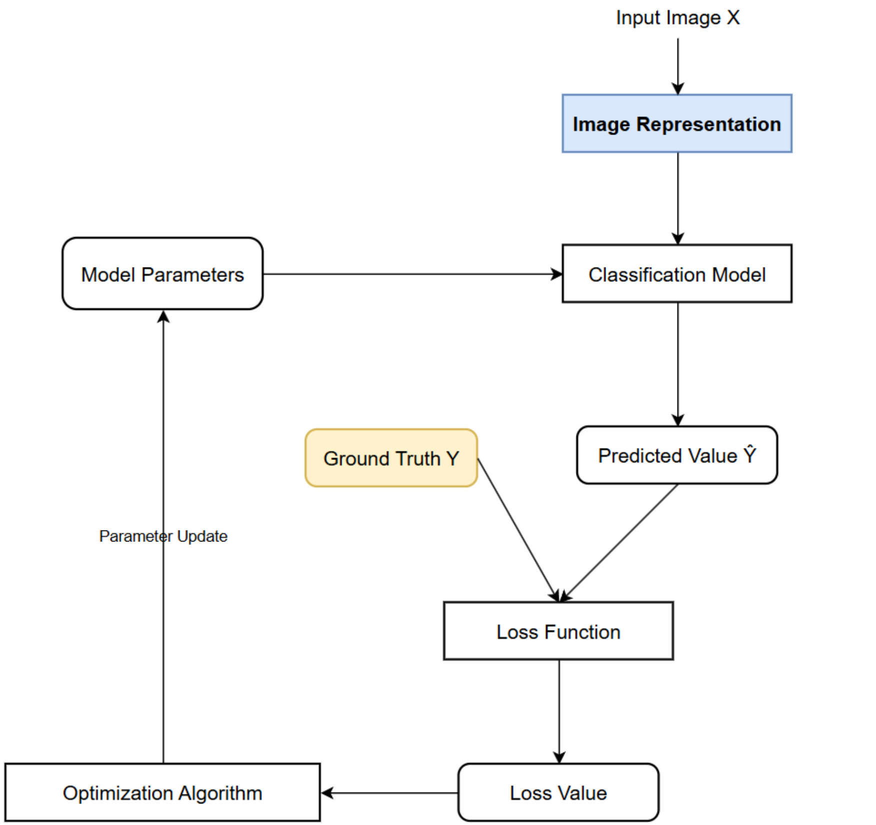


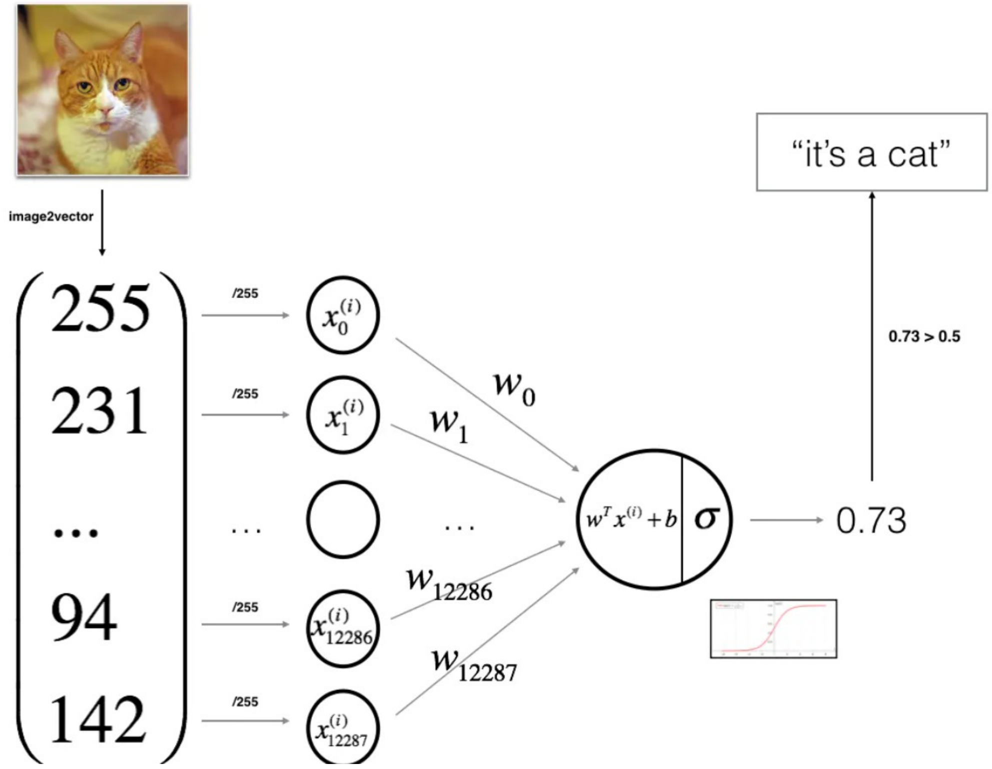
Image Representation

- Pixel Representation
- Global Feature Representation (e.g., GIST)
- Local Feature Representation (e.g., SIFT)

Features + Bag-of-Words Model

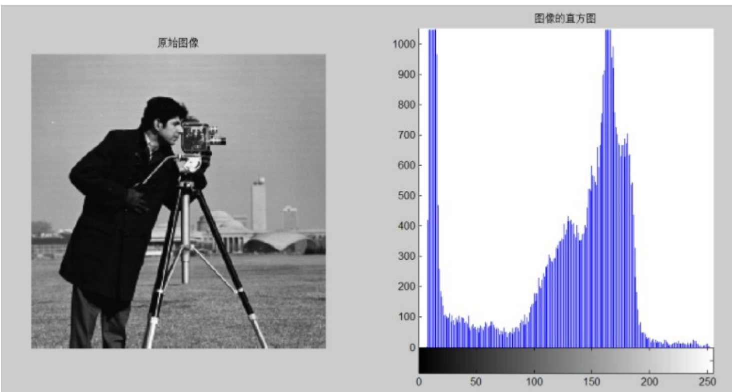
Classifier Design and Learning | Feature Representation

- Pixel Features



Classifier Design and Learning | Feature Representation

Global Feature Extraction – Image Histogram



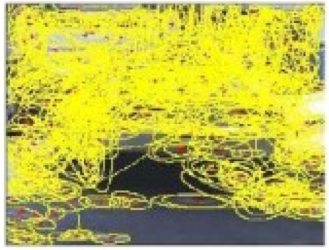
$$H = \{h[1], h[2], \cdots, h[L]\}$$

$$h[k] = n_k / N_i,\ k = 0, 1, \ldots, L-1$$

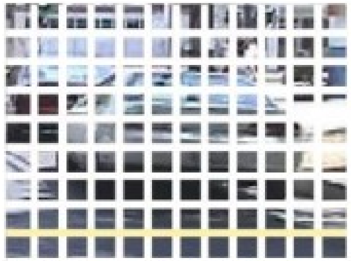
- Calculating a color histogram requires dividing the color space into several small color intervals, each of which becomes a bin of the histogram. This process is called color quantization.
- Then, the color histogram can be obtained by counting the number of pixels whose colors fall into each small interval.
- Suppose a color image contains N pixels, and the color space of the image is quantized into L distinct colors, then the color histogram can be calculated using the following formula.

Classifier Design and Learning | Feature Representation

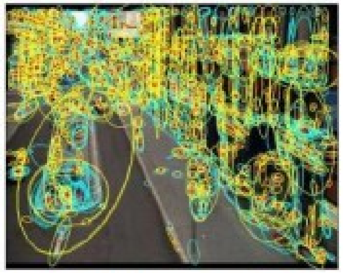
Local Feature Extraction – Sampling Strategy



Interest operators

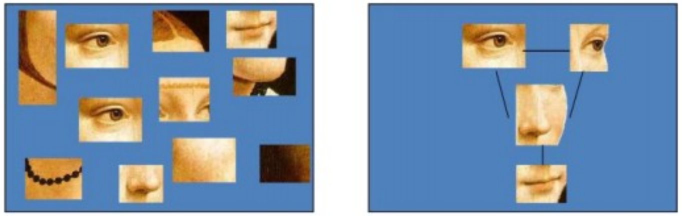


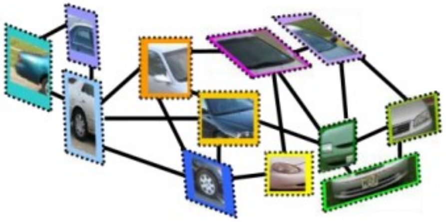
Dense, uniformly



Multiple interest operators

Randomly

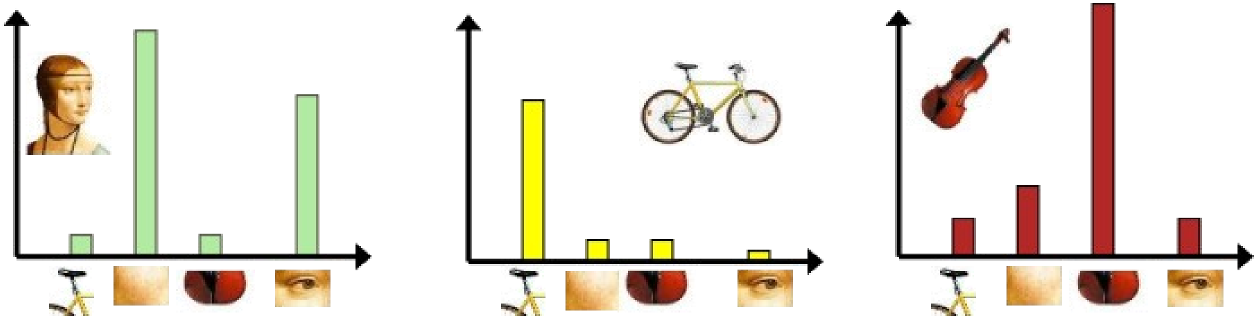




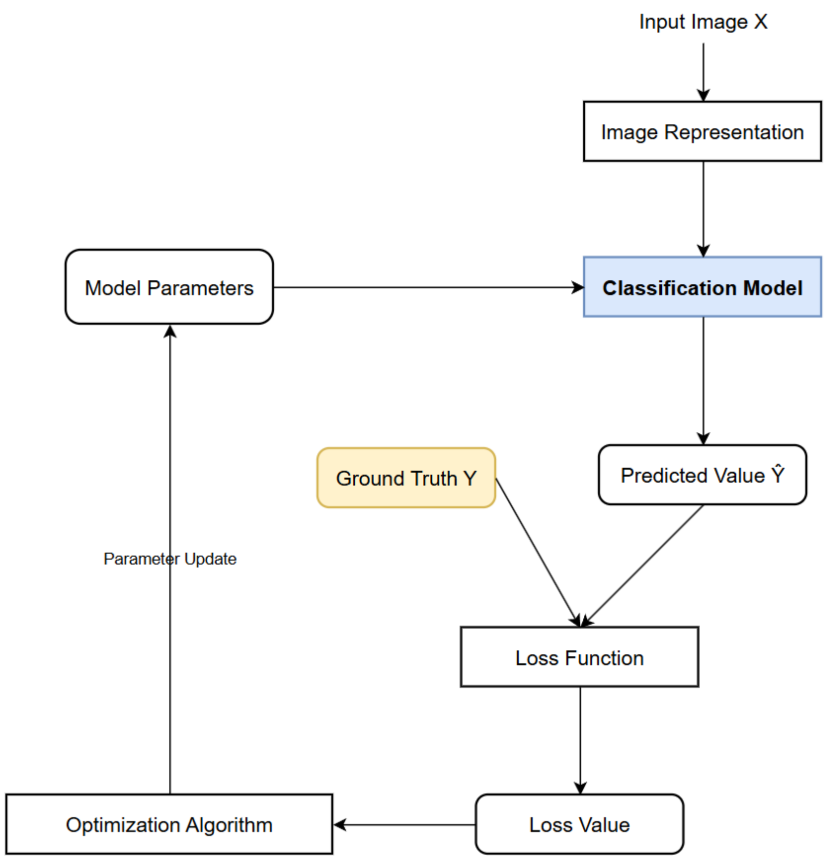
Classifier Design and Learning | Feature Representation

Local Feature Extraction - Bag of Words Model

- Extract Features
- Learn "Visual Words"
- Represent Images by Frequency of "Visual Words"



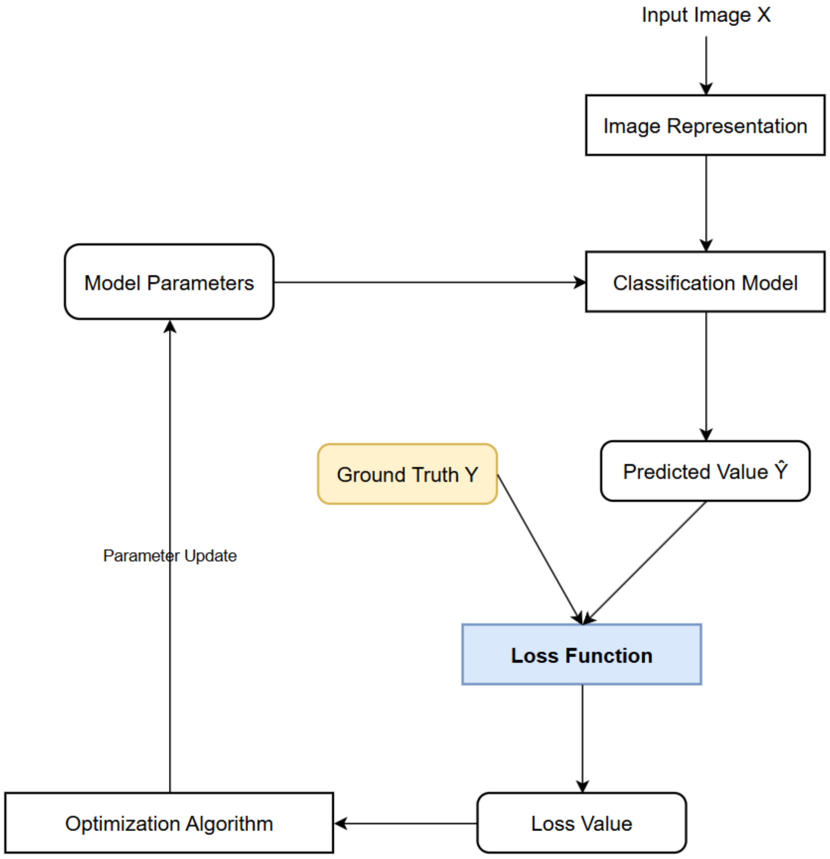
Classifier Design and Learning



Classifiers

- Nearest Neighbor Classifier
- Bayesian Classifier
- Linear Classifier
- Support Vector Machine Classifier
- Neural Network Classifier
- Random Forest Classifier

Classifier Design and Learning

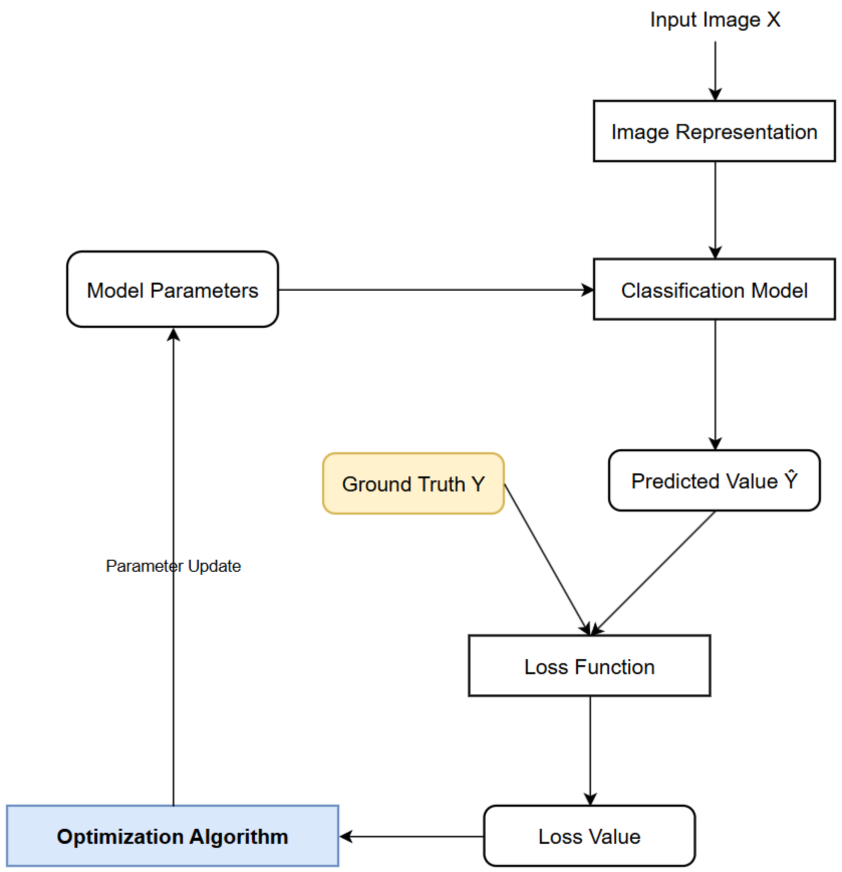


Loss Function

- 0-1 Loss
- Multi-class Support Vector Machine Loss
- Cross-entropy Loss

....

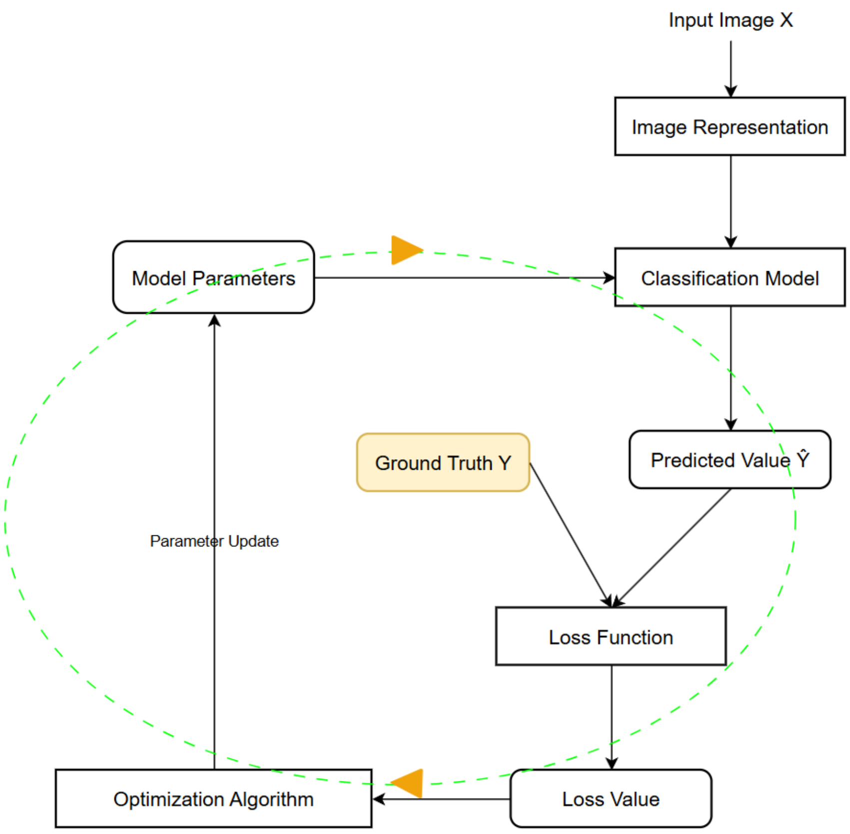
Classifier Design and Learning



Optimization Methods

- First-order Methods
  - Gradient Descent
  - Stochastic Gradient Descent
  - Mini-batch Stochastic Gradient Descent
- Second-order Methods
  - Newton's Method
  - BFGS
  - L-BFGS

Classifier Design and Learning



Training Process

- Dataset Splitting
- Data Preprocessing
- Data Augmentation
- Underfitting and Overfitting
  - Reduce Algorithm Complexity
  - Use Weight Regularization Terms
  - Use Dropout Regularization
- Hyperparameter Tuning
- Model Ensembling

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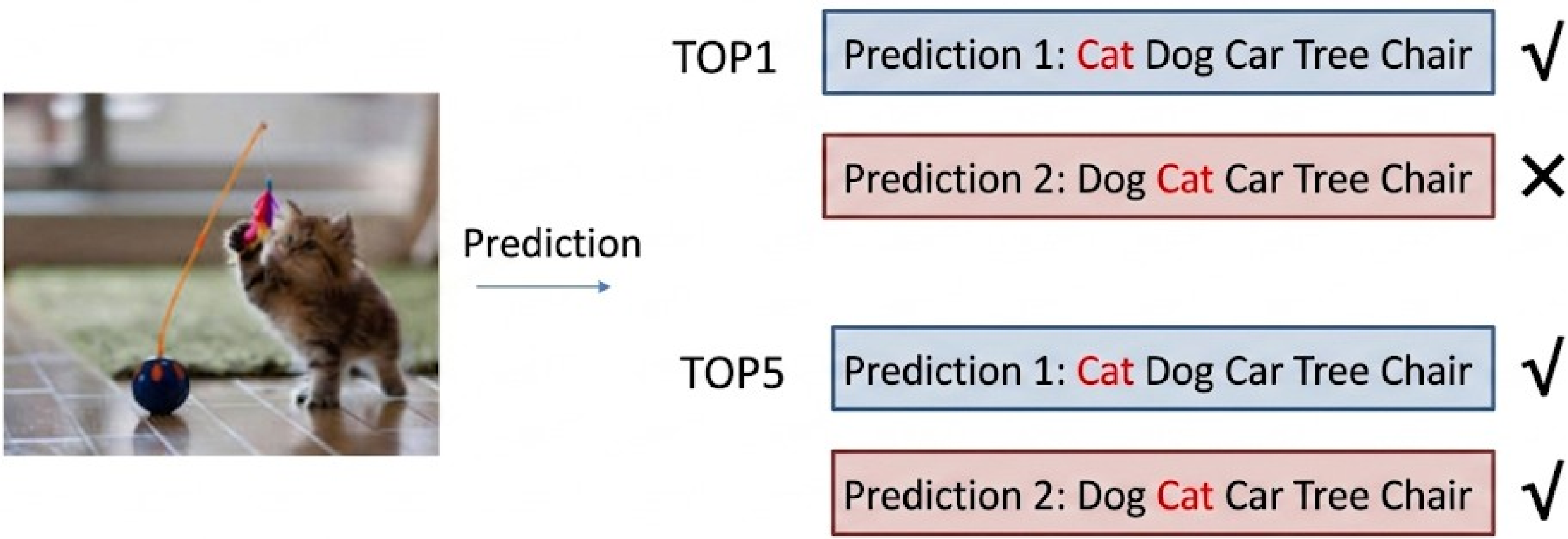
Evaluation Metrics for Image Classification Tasks

- $\text{Accuracy} = \dfrac{\text{Number of correct samples}}{\text{Total number of samples}}$

- Error Rate = 1 - Accuracy

Top-1 Metric and Top-5 Metric



Top-1 Metric and Top-5 Metric